

Read this carefully: For this to be carried out successfully you will need your **MCECS account**. Instructions for getting MCECS accounts (if you don't have one already) are [here](#) or if you need to reset your password are [here](#).

Additional information on how to connect to PostgreSQL can be found [here](#).

Notes

- We will be using **DBeaver** as a platform for accessing and using databases this term.
- Download the community version of [DBeaver](#) .
- If you want to access the database through **psql** then you are free to do so and you can skip to Page 7 for the steps.

Steps to connect through DBeaver:

- ❖ After successfully downloading the DBeaver, click on the 'New Database Connection' button:

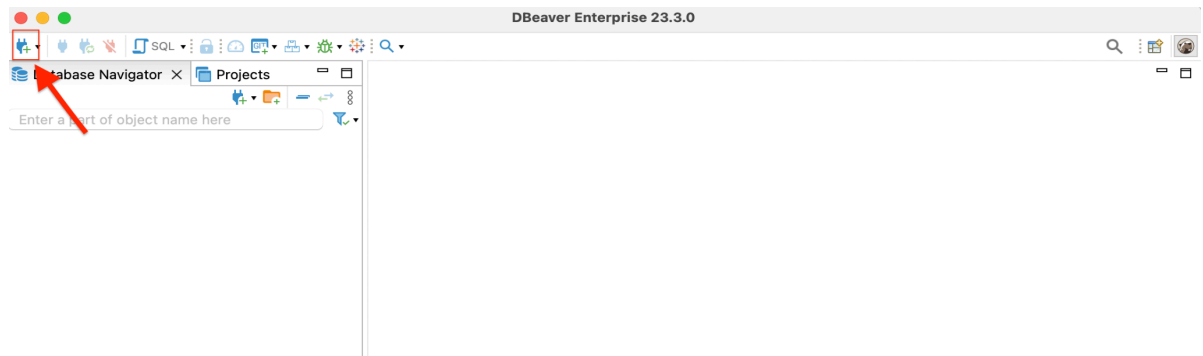


Figure 1

- ❖ Then you will see a box appear in front of the screen with names of different databases. Click on the logo and name with '**PostgreSQL**'(we will be using PostgreSQL this term).

Note: The databases shown in the below image might not be in the version you downloaded, so don't worry about that.

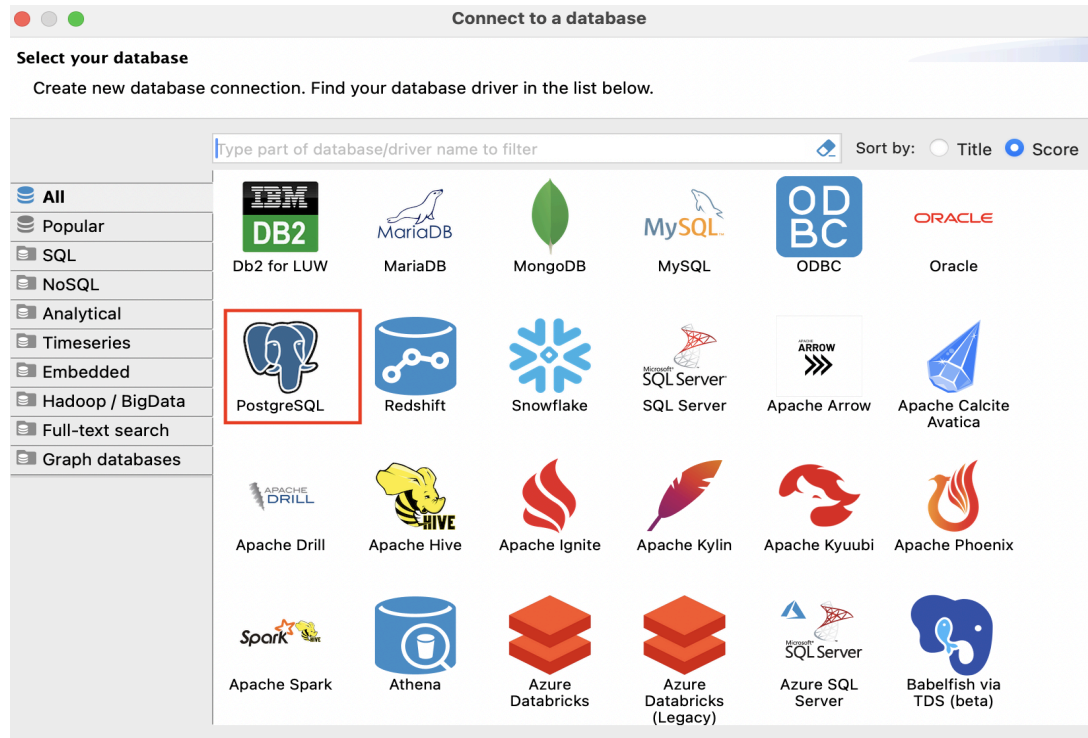
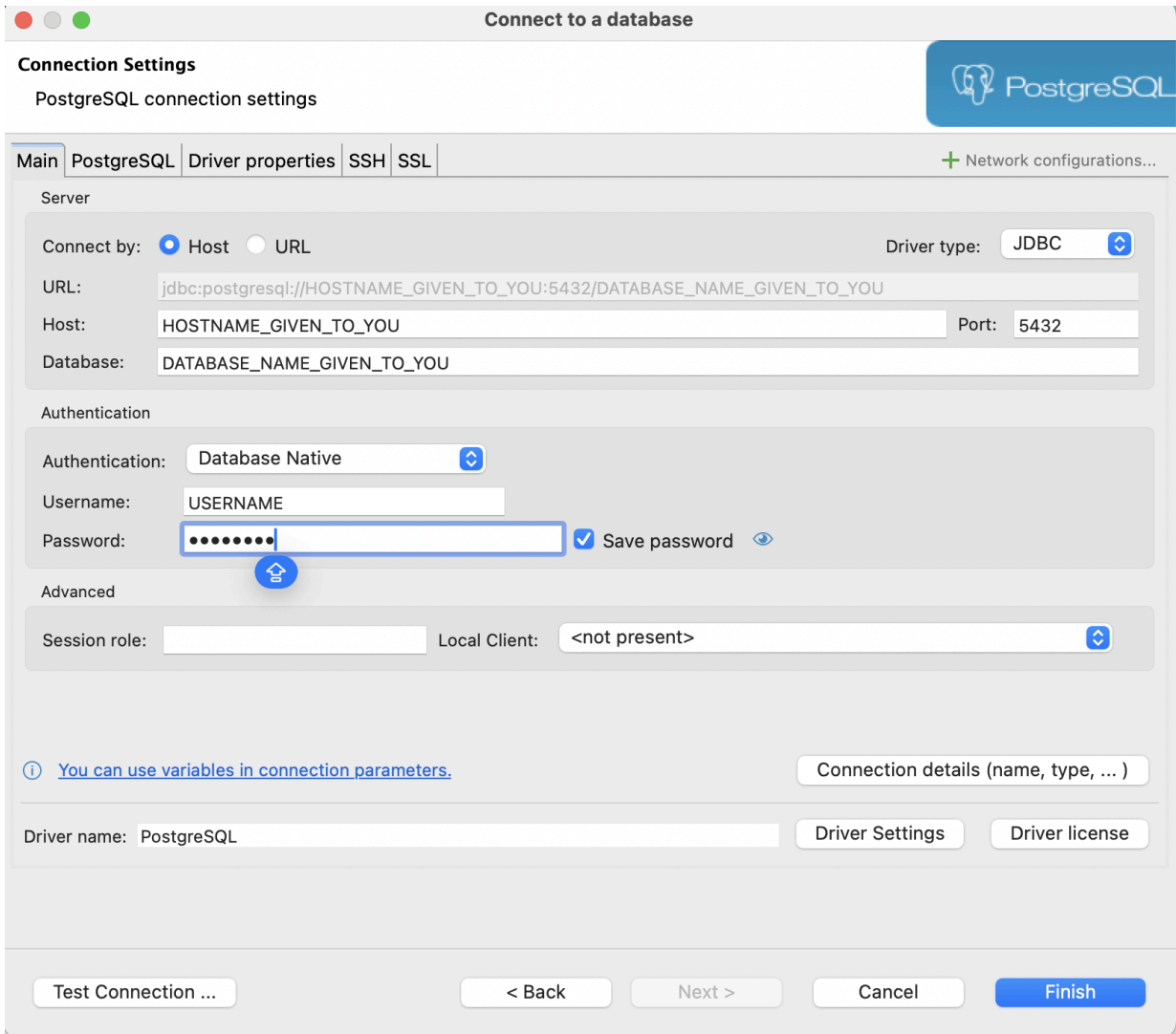


Figure 2

- ❖ Now in the next screen, you should add the **credentials** given to you on your **canvas inbox**, along with the host, database, username and password. Make sure to keep it to yourself and not share with anyone else. The next thing you can do is test connection but there's still one more step to connect successfully so you might get an error message momentarily.



Connect to a database

Connection Settings
PostgreSQL connection settings

Main PostgreSQL Driver properties SSH SSL + Network configurations...

Server

Connect by: ☒ Host ☐ URL Driver type: JDBC

URL: jdbc:postgresql://HOSTNAME_GIVEN_TO_YOU:5432/DATABASE_NAME_GIVEN_TO_YOU

Host: HOSTNAME_GIVEN_TO_YOU Port: 5432

Database: DATABASE_NAME_GIVEN_TO_YOU

Authentication

Authentication: Database Native

Username: USERNAME

Password: ☒ Save password

Advanced

Session role: Local Client: <not present>

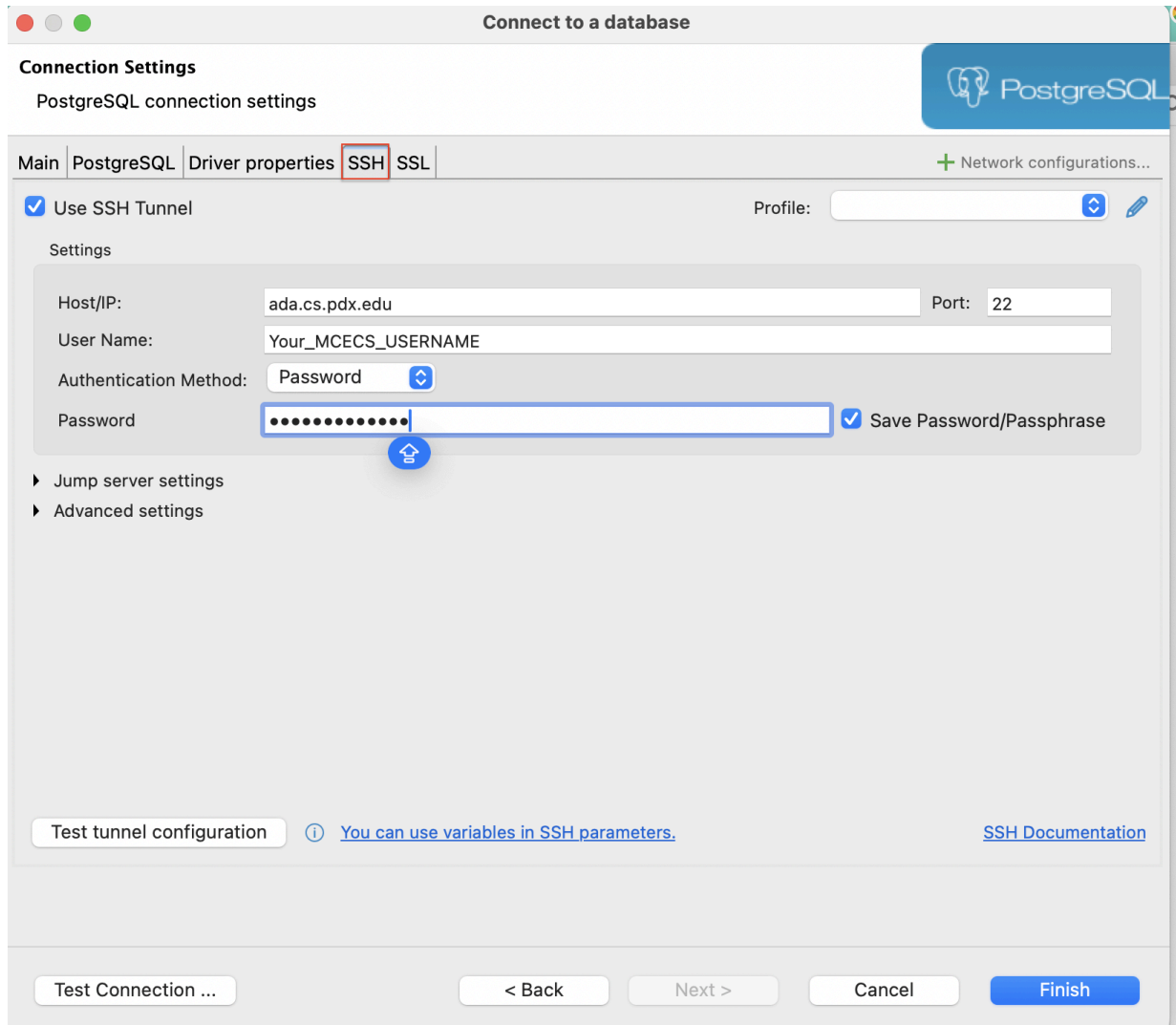
[You can use variables in connection parameters.](#) Connection details (name, type, ...)

Driver name: PostgreSQL Driver Settings Driver license

Test Connection ... < Back Next > Cancel Finish

Figure 3

- ❖ That successful step would be **tunneling**, you will have to **ssh** the tunnel using your **MCECS username** and **password** (if you don't have an MCECS account already follow the instructions highlighted in the first page).



Connect to a database

Connection Settings
PostgreSQL connection settings

Main PostgreSQL Driver properties **SSH** SSL + Network configurations...

☒ Use SSH Tunnel Profile:

Settings

Host/IP: Port:

User Name:

Authentication Method:

Password: ☒ Save Password/Passphrase

▶ Jump server settings
▶ Advanced settings

Test tunnel configuration ⓘ [You can use variables in SSH parameters.](#) [SSH Documentation](#)

Test Connection ... < Back Next > Cancel Finish

Figure 4

- For the **Host/IP** type: ada.cs.pdx.edu or babbage.cs.pdx.edu
- For **UserName** type your MCECS username and for **Password** type your MCECS password. Then click on the **Test tunnel Configuration** and you should see successful message like this:

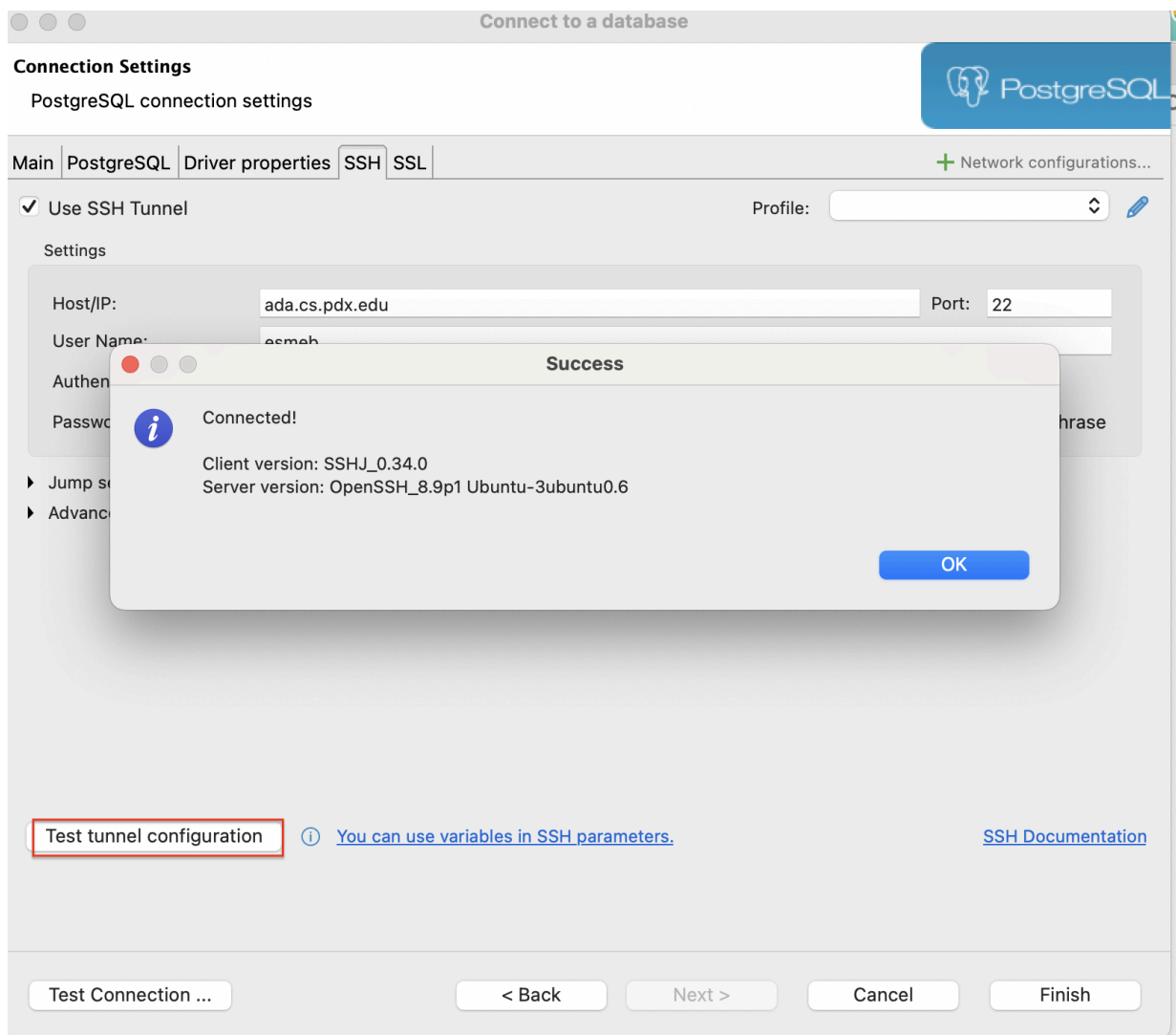


Figure 5

- ❖ Then finally click on the **test connection** button, and you should see a successful message this time.

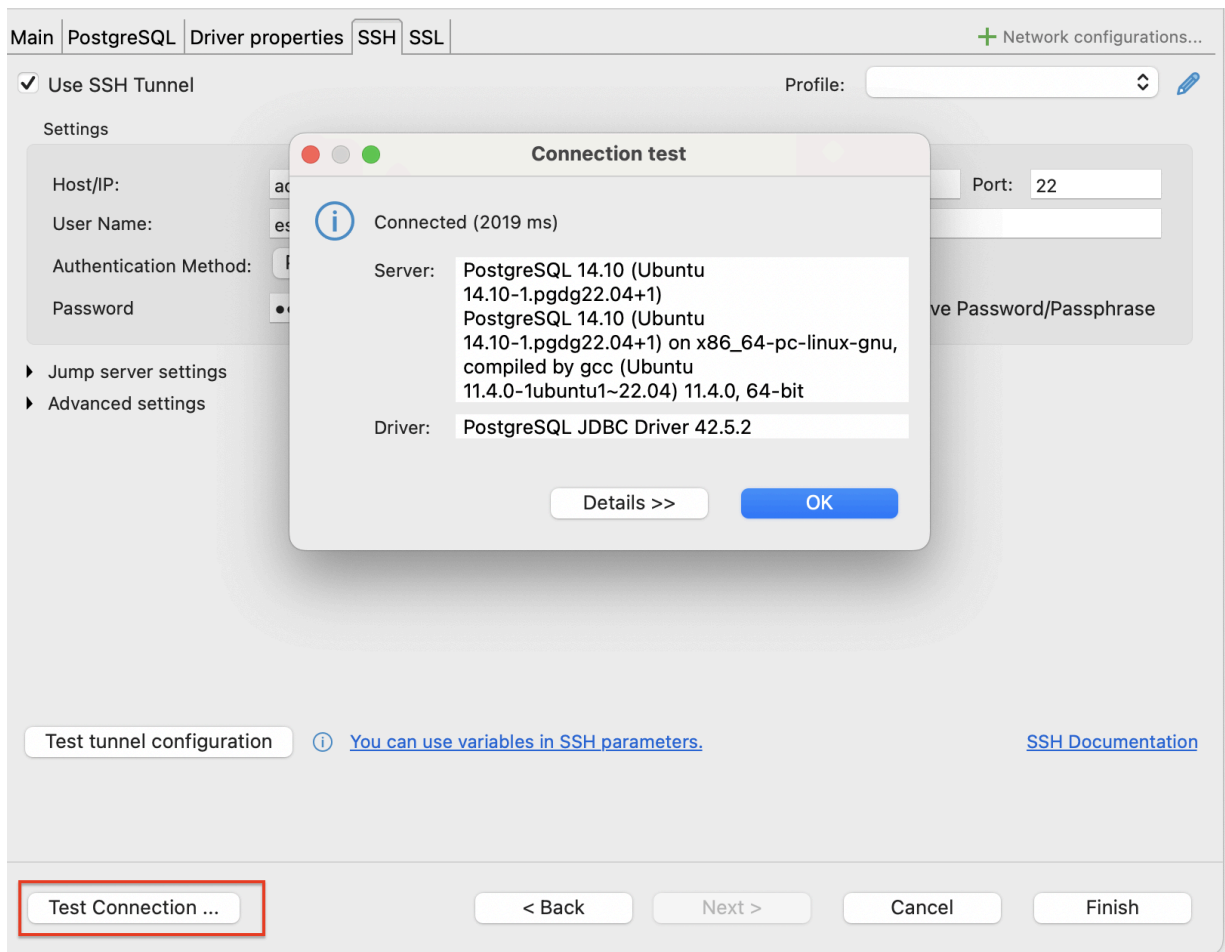


Figure 6

Steps to connect through psql:

a. First, ssh into linux.cs.pdx.edu using your usual MCECS username & password.

b. Then, at the command line, type:

```
psql -h dbclass.cs.pdx.edu -U your_user_name your_user_name
```

- Use the db password provided in the canvas inbox as the password
- **Note:** your_user_name is repeated - the first instance is to indicate your username, the second is actually specifying the name of the database to connect to - your db name is the same as your username

The files from the [Google Drive folder](#) should go in the same directory as the directory in which you are running psql. Be sure to use \copy and not copy.

- Note: copy tries to look for the files on the database machine, which you don't have access to, \copy looks for the files in the directory in which psql is running.